



HealthRAG: Knowledge-Driven Medical QA System

Nguyen Manh Tuan, Tran Ba Cuong, Le Viet Hung, Truong Huu Nhat,
and Phan Duy Hung^(✉) 

FPT University, Hanoi, Vietnam

{tuannmhe171284, cuongtbhe171454, hunglvhe170038,
nhattthhe170351}@fpt.edu.vn, hungpd2@fe.edu.vn

Abstract. A basic understanding of diseases is essential for communities, yet accessing accurate and comprehensible medical information remains challenging, particularly for non-experts in Vietnam. This paper proposes a Retrieval-Augmented Generation (RAG) system for answering disease-related questions, built on a corpus of 610 medical articles collected from youme.vn. The system leverages Halong Embedding, a Vietnamese text embedding model optimized for RAG and fine-tuned on medical data, which boosts retrieval accuracy from 67.44% to 90.57% (Top@10) at dimension 128, and from 55.69% to 79.23% (Top@3) at dimension 256. Experiments demonstrate strong retrieval effectiveness, achieving Recall@10 of 90.57% and NDCG@10 of 0.76902, with accuracy improving by more than 23% after fine-tuning. The main contribution lies in developing a pioneering Vietnamese medical Q&A system that combines multiple retrieval methods—semantic embedding, keyword search (BM25), and structured reasoning via a medical knowledge graph (KG)—to deliver fast and reliable access to information. The strongest performance was achieved with a hybrid configuration (Hybrid-D) integrating all three methods, reaching Accuracy@10 of 90.11%, Accuracy@1 of 72.70%, MRR@3 of 77.70%, and NDCG@10 of 81.67%, significantly outperforming any single component. Finally, Google Gemini is employed as the large language model to generate natural, context-aware answers. The resulting system demonstrates the feasibility of building an end-to-end Vietnamese medical QA platform that effectively integrates hybrid retrieval with advanced generation, thereby improving healthcare information access in low-resource language settings.

Keywords: RAG · medical Q&A · Embedding Model · KG · keyword search

1 Introduction

In today's digital age, online health information shapes public understanding but often results in information overload, inconsistent accuracy, and complex jargon that challenge non-specialists. This issue is particularly critical in Vietnam, where over 75% of internet users seek medical advice online, yet face a shortage of high-quality Vietnamese resources and generally low levels of medical literacy. Traditional search engines

frequently return irrelevant or overly technical content, while Large Language Models (LLMs), although conversational, are prone to hallucinations that can jeopardize healthcare decisions. To address this gap, we propose a Vietnamese medical Question Answering (QA) system that combines the structured accuracy of retrieval methods with the generative fluency of LLMs. By leveraging curated data and multiple retrieval techniques, the system delivers reliable, contextualized answers tailored to the Vietnamese medical domain.

Medical Question Answering has advanced significantly with the rise of deep learning and LLMs, evolving from rule-based approaches to transformer-based models such as BERT, BioBERT, and ClinicalBERT [1], which have improved entity recognition and clinical text comprehension. However, LLMs are prone to hallucinations—producing incorrect or unverifiable responses [2, 3]—which poses serious risks in healthcare. This limitation has driven a shift toward Retrieval-Augmented Generation, which integrates retrieval with generative modeling to ground answers in verifiable data, thereby reducing errors while improving accuracy and explainability [4, 5]. The concept of RAG for knowledge-intensive NLP tasks was first introduced by Lewis et al. [6], later refined with Fusion-in-Decoder (FiD) [7], and subsequently applied in domain-specific systems such as BioRAG [8] and MedRAG [9]. These systems consistently outperform end-to-end generative models in recalling medical facts.

A critical factor in RAG's effectiveness is its retrieval component. Sparse techniques such as BM25 [10] excel at handling keyword-dense queries, while dense retrievers like Sentence-BERT and E5 [11, 12] capture semantic nuances through embeddings. Hybrid retrieval methods [13] combine these approaches to achieve greater robustness, and Knowledge Graphs (KGs) [14]—such as UMLS and SNOMED-CT—enable structured reasoning in medical QA. Despite these advancements, Vietnamese medical QA remains underexplored due to limited domain-specific datasets, the scarcity of pre-trained models, and inherent linguistic complexities. Existing models like PhoBERT [15], while effective for general NLP tasks, are not specialized for the healthcare domain.

This paper makes several key contributions. First, it presents a pioneering Vietnamese medical RAG system for accurate and trustworthy disease-related question answering, integrating Halong Embedding (a fine-tuned Vietnamese text embedding model), a large language model (Google Gemini), and a custom-built medical knowledge graph. Second, it introduces a curated dataset comprising more than 600 Vietnamese medical articles from *youme.vn* and over 5,000 manually crafted medical QA pairs, which are used for system fine-tuning and evaluation. Third, it evaluates the effectiveness of hybrid retrieval strategies that combine semantic embedding, keyword-based retrieval (BM25), and graph-based retrieval, demonstrating significant improvements in both precision and recall. The results show that the hybrid approach notably outperforms any individual retrieval method.

2 Methodology

2.1 Model Architecture

In this subsection, the architecture of our HealthRAG system is outlined, providing a comprehensive overview of its components and data flow, as illustrated in Fig. 1. The system is designed as an end-to-end Retrieval-Augmented Generation framework [6], focusing on delivering accurate and contextually relevant answers to Vietnamese medical queries, with integrated multimodal capabilities.

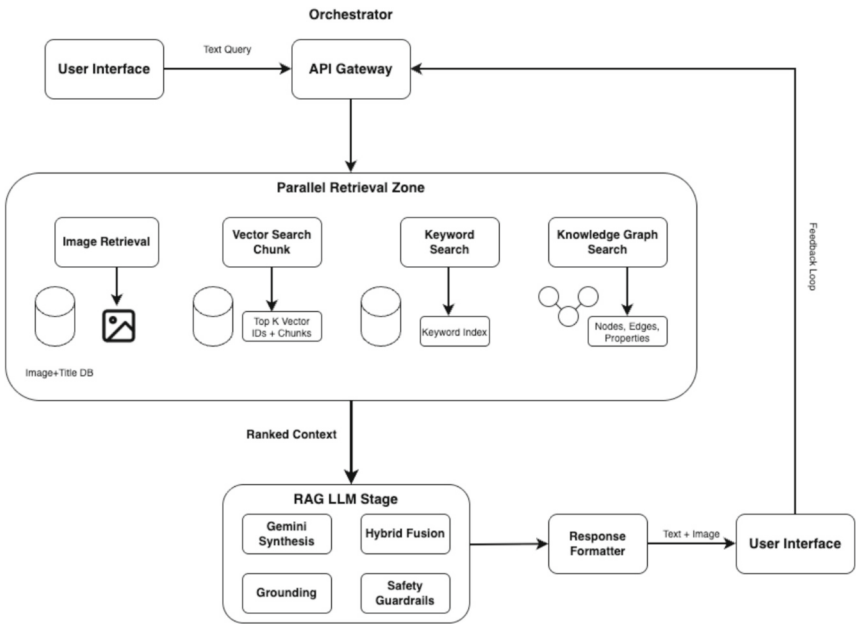


Fig. 1. Overall Pipeline of the Vietnamese Medical RAG System

The Retrieval Module is the heart of HealthRAG’s information gathering, featuring a Hybrid Retrieval Module that conducts parallel searches across three knowledge sources. Dense Retrieval uses semantic similarity with Halong Embedding [12] to search the “DB Text+Title” database, excelling at contextual relevance. Sparse Retrieval employs the BM25 algorithm [10] to match lexical terms in “DB Text” and “Keyword Index,” ensuring high precision. Knowledge Graph Retrieval queries the Graph Database, leveraging interlinked Nodes, Edges, and Properties [14] for structured medical insights. The Context Fusion step uses Reciprocal Rank Fusion (RRF) [13] to merge and re-rank results, reducing biases and producing a unified, relevant context set. This is then fed into the Generative Module, powered by Google Gemini LLM [16], which synthesizes coherent answers, handles multimodal data, and applies prompt engineering and post-processing to ensure accuracy and safety in medical outputs.

2.2 Data Collection and Preprocessing

The RAG system’s knowledge base is built from 610 medical articles scraped from youme.vn, complemented by 1882 associated images. A rigorous preprocessing pipeline cleaned this multimodal raw data, involving HTML tag/script removal, content filtering, character normalization, and whitespace elimination. Initial data analysis, including visualizing the distribution of article lengths (Fig. 2), was performed to inform optimal chunking strategies. For efficient retrieval, cleaned articles underwent distinct preparation processes. For Dense Retrieval, articles were segmented into overlapping chunks (200 words, 50 words overlap) and transformed into vector embeddings via Halong Embedding, indexed in FAISS. For Sparse Retrieval, passages were preprocessed and indexed using the BM25 algorithm [10]. A domain-specific Knowledge Graph was constructed through medical Named Entity Recognition (NER) and Relation Extraction (RE), identifying entities and relationships from the text [14].

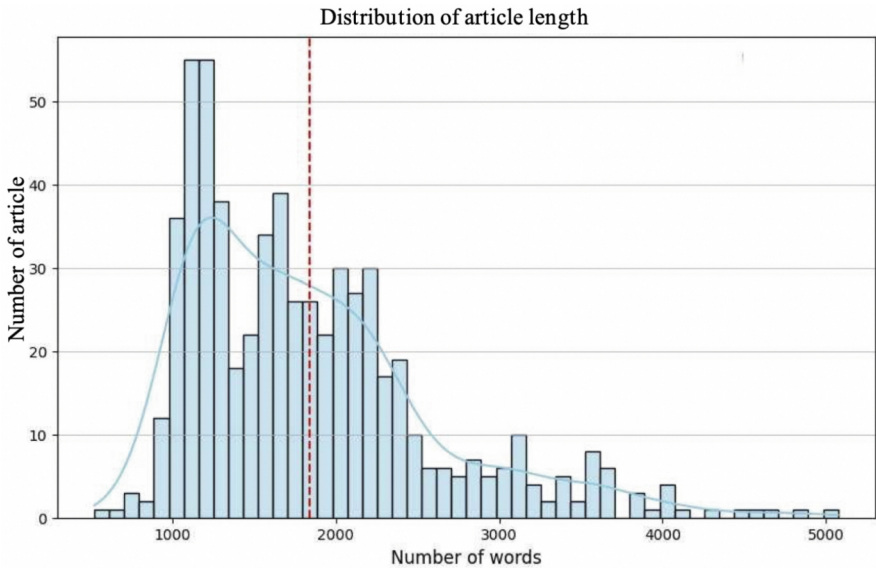


Fig. 2. Distribution of Article Lengths in the Medical Corpus

2.3 Halong Embedding Model Training

The Halong Embedding model, a Vietnamese Sentence Transformer [12], is central to our dense retrieval. Its development involved two stages. In the pre-training stage, a multilingual base model (intfloat/multilingual-e5-base) was used to acquire general Vietnamese language understanding. Subsequently, fine-tuning on the Medical QA Dataset specialized the model for the medical domain, employing a Siamese network architecture [11] on an in-house dataset of approximately 100,000 QA pairs. This fine-tuning optimized the embedding space for semantic similarity by employing a composite loss function that combined Multiple Negative Ranking Loss (MNRL) and Matryoshka Loss.

2.4 Retrieval Strategy

We employ a multi-faceted retrieval strategy that combines semantic understanding with precise keyword matching. Embedding-based search leverages the fine-tuned Halong Embedding [12] to identify semantically similar document chunks via cosine similarity search on a FAISS index. Keyword-based search applies the BM25 algorithm [10] on an inverted index for lexical matching. To maximize overall retrieval performance, a hybrid retrieval mechanism fuses the independently ranked lists from both methods using Reciprocal Rank Fusion (RRF) [13]. This robust approach merges diverse rankings without requiring score calibration, thereby producing a highly optimized Top-N context for the LLM.

2.5 Knowledge Graph Construction and Integration

A domain-specific knowledge graph was constructed from the cleaned article corpus [14]. This process involved entity and relation extraction using custom Vietnamese medical NER and RE techniques to identify entities (e.g., diseases, symptoms) and their relationships. The KG enhances both retrieval and generation: during retrieval, relevant subgraphs are extracted and serialized into text to augment context; during generation, the KG provides factual grounding, reducing hallucinations by serving as a structured reference.

2.6 Gemini LLM for Answer Generation

The retrieved and fused context is fed into the Google Gemini LLM [16] to generate natural, coherent, and context-aware answers. Prompt engineering carefully structures the inputs (system prompt, retrieved context, user query) to guide Gemini’s behavior. During generation, post-processing and safety mechanisms are applied. This stage improves output readability and incorporates essential checks such as hallucination detection [2, 4], harmful content filtering, and the inclusion of necessary disclaimers, thereby ensuring the delivery of reliable and safe medical information.

3 Experiments and Results

3.1 Datasets and Evaluation Metrics

We constructed two primary datasets to support the development and evaluation of the RAG system (Table 1). The Medical Article Corpus comprises 610 medical articles meticulously scraped from youme.vn, each focused on a distinct disease. Following rigorous cleaning and preprocessing (noise filtering, typo correction, terminology standardization), the corpus contains 1,122,536 words, with an average article length of approximately 1,840 words. In addition, a manually curated dataset of 5,215 QA pairs was generated, with 8–9 questions crafted for each of the 610 articles, covering symptoms, treatments, and general notes. This complete dataset was employed for both fine-tuning Halong Embedding and evaluating the system’s performance, without being partitioned into separate training and test subsets.

Table 1. Dataset Statistics

Metric	Value
Number of Articles	610
Total Words	1,122,536
Number of QA Pairs	5215
Average Words per Article	1840

We primarily evaluated the system based on retrieval performance. The reported metrics reflect the combined effectiveness of all retrieval components on the 5,215 QA pairs. The retrieval metrics employed include: Accuracy@k, measuring the proportion of queries where the correct document is retrieved within the top k results; Precision@k, assessing the fraction of relevant documents among the top k retrieved results; Recall@k, evaluating the proportion of relevant documents successfully retrieved within the top k results; NDCG@10 (Normalized Discounted Cumulative Gain at $k = 10$), quantifying the ranking quality of retrieved documents; MRR@10 (Mean Reciprocal Rank at $k = 10$), focusing on the rank position of the first relevant document; and MAP@100 (Mean Average Precision at $k = 100$), capturing overall retrieval precision across queries.

3.2 Experiment Settings

Halong Embedding was fine-tuned on the entire combined dataset (5,215 QA pairs) to optimize medical text retrieval performance. Training was conducted for approximately three epochs (2.96933), resulting in a final loss of 1.0182. The process required about 93 min (5,585.30 s) over 243 global steps.

The hardware used for training and evaluation consisted of two NVIDIA T4 GPUs (16 GB VRAM each) and 32 GB RAM on Kaggle, with Python 3.8 and PyTorch as the software environment. The RAG system’s retrieval component integrated a TF-IDF-based keyword search to complement the embedding-based method. In addition, a knowledge graph was constructed from disease–symptom–treatment relationships to further enhance semantic understanding.

3.3 Halong Embedding Performance (Pre- vs. Post-fine-Tuning)

We evaluated Halong Embedding across five dimensions (64, 128, 256, 512, 768) to assess retrieval performance before and after fine-tuning. Table 2 summarizes these results, highlighting the direct impact of fine-tuning. For example, the dim_256 configuration showed a notable improvement in Accuracy@1, increasing from 0.4422 to 0.6374. Overall, fine-tuning consistently enhanced performance across all dimensions. In particular, dim_64 achieved a substantial +89.64% gain in Accuracy@1 (from 0.3185 to 0.6044), while dim_768 improved Accuracy@10 by +20.30% (from 0.7459 to 0.8974). Among all configurations, dim_256 demonstrated the best overall performance, with Accuracy@10 improving by +27.15% (from 0.7103 to 0.9028).

Table 2. Pre- and Post-Fine-tuning Retrieval Performance (by Dimension)

Dimension	Phase	Accuracy@1	Accuracy@3	Accuracy@10
64	Pre	0.3185	0.4547	0.5827
64	Post	0.6044	0.7661	0.8901
128	Pre	0.4019	0.5569	0.6744
128	Post	0.6313	0.7900	0.9057
256	Pre	0.4422	0.6008	0.7103
256	Post	0.6374	0.7923	0.9028
512	Pre	0.4587	0.6309	0.7377
512	Post	0.6328	0.7837	0.8993
768	Pre	0.4666	0.6405	0.7459
768	Post	0.6334	0.7862	0.8974

3.4 Keyword Search Performance (BM25)

Rewrite this base on new numeric As a robust non-embedding baseline, BM25 was applied to the Combined Dataset. Its performance was assessed across top-k thresholds ($k = 1, 3, 5, 10$). Table 3 shows that accuracy and recall steadily increased with larger k (reaching 91.25 values also improved, indicating relevant documents were frequently located early in the result list. BM25 demonstrated strong baseline performance, justifying its use as a complementary retrieval signal.

Table 3. BM25 Retrieval Performance

Metric	@1	@3	@5	@10
Accuracy	0.4769	0.7082	0.7731	0.8278
Precision	0.4769	0.2361	0.1546	0.0828
Recall	0.4769	0.7082	0.7731	0.8278
NDCG	0.4769	0.6158	0.6126	0.6006
MRR	0.6114	–	–	–

Table 4. Entity-Level Evaluation

Metric	Score
Precision	1.0000
Recall	0.8110

(continued)

Table 4. *(continued)*

Metric	Score
F1-score	0.8956
Exact Match Accuracy	0.5738
Avg. Jaccard Score	0.8125
Entity Coverage	0.6633

3.5 Knowledge Graph Evaluation

The Knowledge Graph was evaluated using a dataset of 5,215 Vietnamese medical questions, with its entity-level performance detailed in Table 4, demonstrating exceptional results. The evaluation revealed perfect precision at 1.0000, indicating that all identified entities were correct, alongside a robust recall of 0.8110, signifying that a significant portion of relevant entities were successfully retrieved. The F1-score, which balances precision and recall, stood at 0.8956, reflecting a strong overall performance. Additionally, the exact match accuracy was recorded at 57.38%, suggesting a moderate success rate in achieving precise entity matches. These metrics collectively underscore the effectiveness of the Knowledge Graph in handling complex medical queries, supported by the integration of advanced retrieval strategies and a well-curated dataset.

3.6 Hybrid Search Performance

To optimize retrieval, we tested hybrid strategies combining Halong Embedding, BM25, and KG-driven retrieval across four configurations on a 5,215 QA dataset: Hybrid-A (embeddings and BM25), Hybrid-B (KG and embeddings), Hybrid-C (KG and BM25), and Hybrid-D (fusing all three via majority voting and weighted rank aggregation). Evaluated with Accuracy@k, MRR@k, and NDCG@k ($k = 1, 3, 5, 10$), Hybrid-D outperformed others, achieving Accuracy@1 of 72.63% and NDCG@10 of 81.67%, showcasing enhanced precision and ranking quality due to the complementary strengths of each source.

4 Conclusion and Future Works

HealthRAG is a Vietnamese medical question-answering system that leverages a hybrid retrieval strategy, combining Halong Embedding, BM25, and Knowledge Graph to provide accurate and reliable responses. This approach demonstrates strong effectiveness in addressing both linguistic and domain-specific challenges in the Vietnamese medical context while ensuring scalability through efficient processing. By integrating multiple retrieval methods, HealthRAG enhances the reliability of medical information delivery and supports more trustworthy healthcare knowledge access.

Future enhancements include integrating multimodal outputs with annotated images from youme.vn for better user comprehension, implementing time-aware updates to keep

the corpus current, and personalizing responses based on user profiles. Additionally, applying Graph Neural Networks for deeper reasoning, deploying HealthRAG in real-world settings with user feedback, and optimizing computational efficiency through lower-dimensional embeddings and cloud APIs will ensure scalability and accessibility. Strengthening safety mechanisms, such as improved hallucination detection and ethical compliance, will further enhance trustworthiness.

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